Introduction

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Motivation

- Signals and systems deal with the **fundamental tools** for engineers in a wide spectrum of fields.
- Signals and systems are **two** of the most **important and immensely used areas** of mathematics.
- **Extermely powerful frameworks** to analyse and synthesise signals and systems.
- **Characterises** in various ways the various signals and systems that are in real-world. How to describe them, measure them, control them, etc.

The course of this subject will cover:

- the **mathematical framework**
  - core theoretical concepts
- **characterisitics** aspects
  - various properties
- **applications** in various fields of signals and systems
  - solving problems related to signal processing and systems

The course of this subject will be divided roughly into two parts:

- **theory**
  - core theory, properties, algorithms and implementation aspects
- **tutorial**
  - solving problems related to signal processing and system science, design, etc.

- **Reference materials:**
  - Text book:
    - Signals and Systems: Alan V. Oppenheim, Alan S. Willsky with S. Hamid Nawab, Pearson Education, India, 2nd Edition
  - Reference books and articles:
    - Principles of Linear Systems and Signals: B. P. Lathi, Oxford: 2nd Edition, 2009

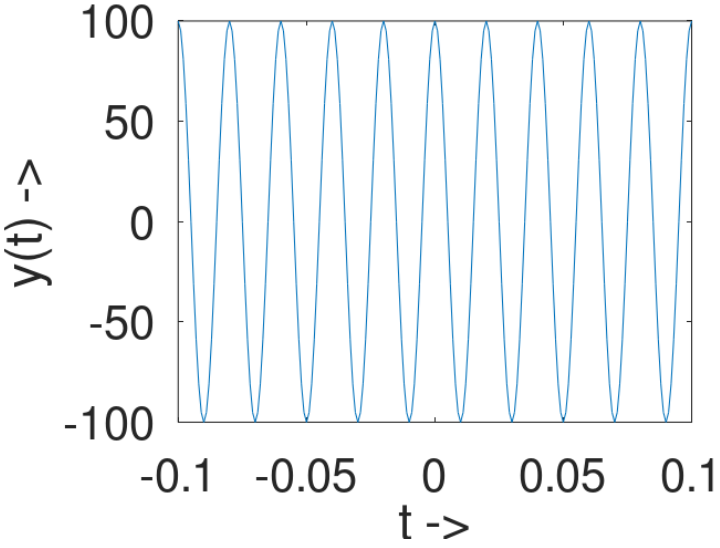
Signals

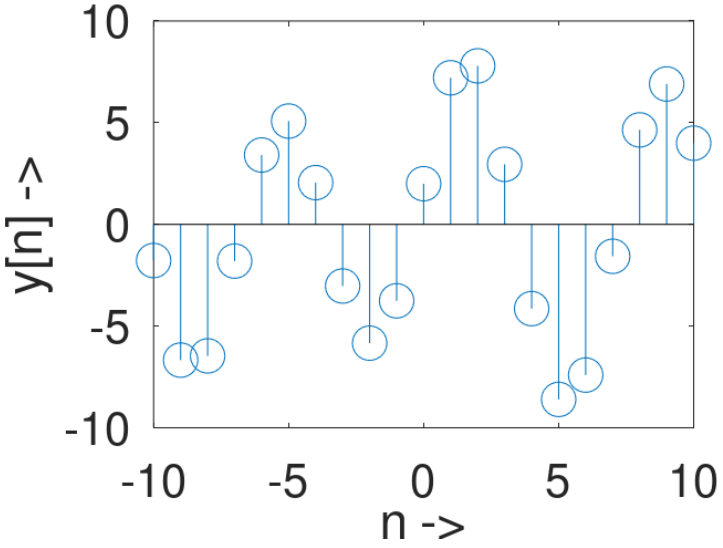
- **Signals**
  - are mathematical functions of a number of independent variables that are usually called indices.
    - e.g. $f(t) = 100\sin(100\pi t)$, $f(x, y) = 2x + y$
    - probably recently come across voltage signals, current signals, etc.
    - but, they are much more - position, velocity, momentum, volume, flow rate, etc.
    - multi-dimensional(multi-indexed) - 2D photographic images, hyperspectral images, etc.
    - financial, geographical, network (social networks and any other), etc.
    - scalar signals, vector signals, etc.

Signals

- **Classification of Signals**
  - **Dimensionality**:
    - One-dimensional: one independent variable. (e.g. $f(t)$ defines ambient temperature)
    - Multi-dimensional: Two or more independent variables. (e.g. $f(x, y)$ defines brightness of sky (How?))
  - Type of the **domain/indices**:
    - Continuous: Independent variables are continuous and can be mapped to an interval in the real line ($\mathbb{R}$).
    - Discrete: Independent variables are discrete and can be mapped to a subset of the integers ($\mathbb{Z}$).
  - **Range**:
    - Analog: Continuous-valued (real/complex) signals
    - Digital: Discrete-value signals.

Signals





Signals-Energy

- Consider a continuous-time signal $x(t)$ defined over real time t.
- Energy of the signal, $E_x(t_1, t_2)$ between times t_1 and t_2 is defined as
$$E_x(t_1, t_2) = \int_{t_1}^{t_2} |x(t)|^2 \, dt.$$
- Consider a discrete-time signal $x[n]$ defined over discrete time index n.
- Energy of the signal, $E_x(n_1, n_2)$ between times n_1 and n_2 is defined as
$$E_x(n_1, n_2) = \sum_{n=n_1}^{n_2} |x[n]|^2 .$$

Signals- Average Power

- Consider a continuous-time signal $x(t)$ defined over real time t.
- Average Power of the signal, $P_x(t_1, t_2)$ between times t_1 and t_2 is defined as

$$P_x(t_1, t_2) = \frac{E_x(t_1, t_2)}{(t_2 - t_1)}$$

- Consider a discrete-time signal $x[n]$ defined over discrete time index n.
- Average Power of the signal, $P_x(n_1, n_2)$ between times n_1 and n_2 is defined as

$$P_x(n_1, n_2) = \frac{E_x(n_1, n_2)}{(n_2 - n_1 + 1)}.$$

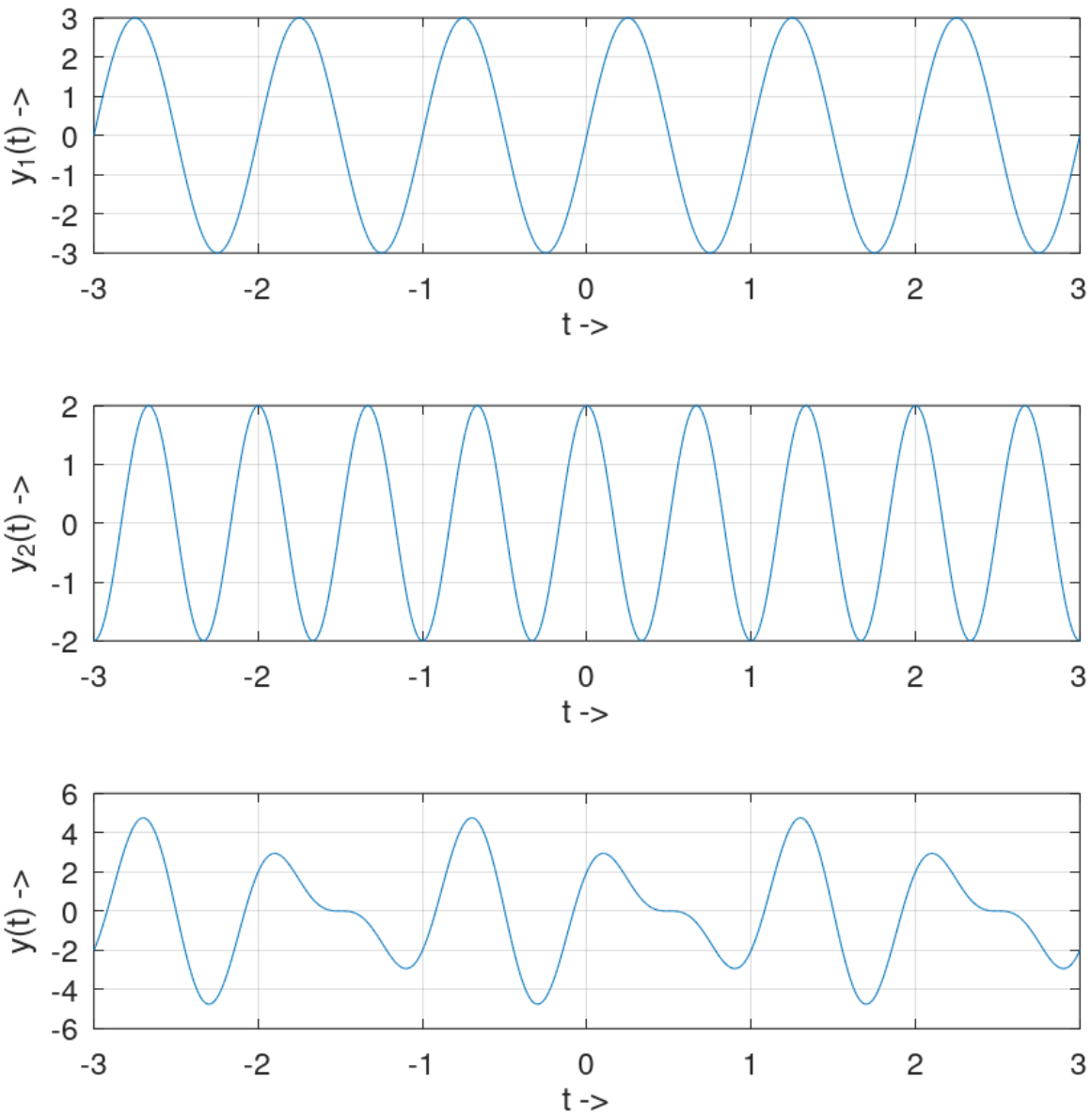
Signals- Average Power

- Energy signals - Signals having finite energy.
- Power signals - Signals having finite average power.
- Other than above - Signals that are neither energy signals nor power signals.

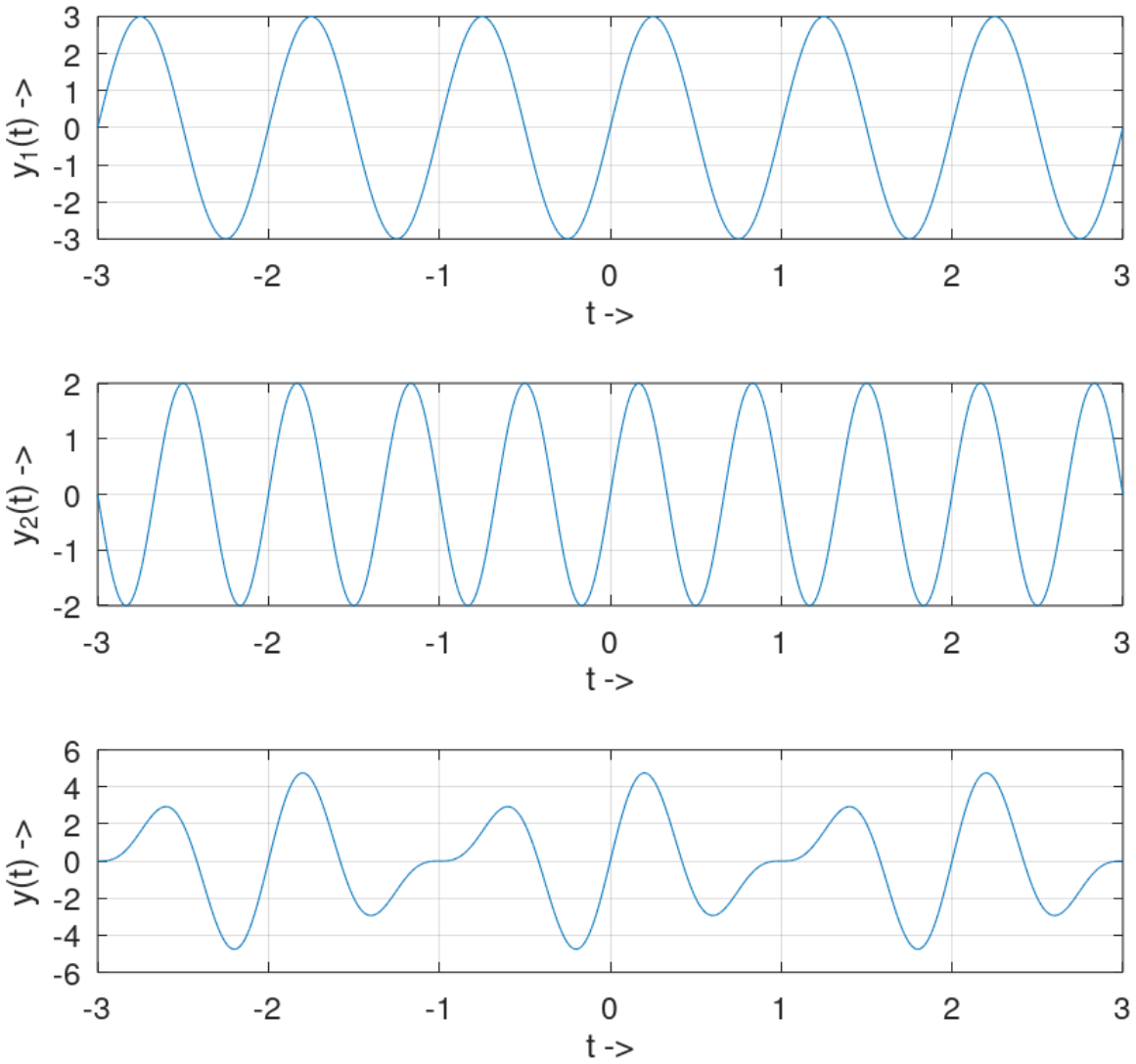
Signals

- **Symmetry**
  - Even signals: $x(-t) = x(t)$, $x[-n] = x[n]$.
  - Odd signals: $x(-t) = -x(t)$, $x[-n] = -x[n]$.
- **A simple signal analysis**
  - Given a signal $y(t)$, we can decompose it as a sum of an even signal $e(t)$ and an odd signal $o(t)$
    - $y(t) = e(t) + o(t)$
    - $e(t) = \frac{1}{2}(y(t) + y(-t))$ and $o(t) = \frac{1}{2}(y(t) - y(-t))$.
    - can do much more interpretation.

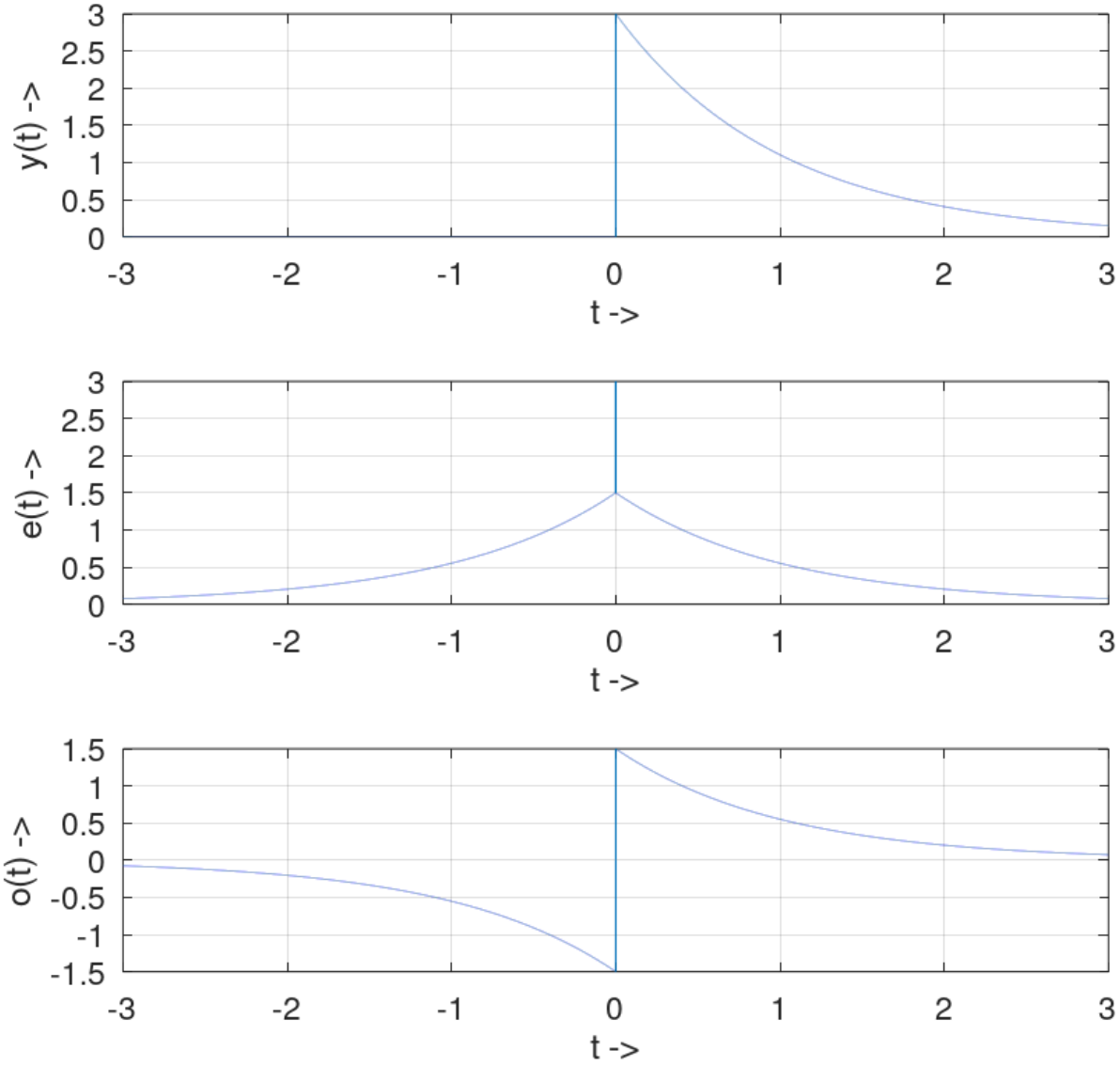
Signals - Simple Synthesis



Signals - Simple Synthesis



Signals - Simple Analysis



- **Conjugate symmetry**
  - Conjugate Symmetry: $x(-t) = x^*(t)$, $x[-n] = x^*[n]$.
  - Conjugate Anti-symmetry: $x(-t) = -x^*(t)$, $x[-n] = -x^*[n]$.
  - Examples?
  - How they related to odd-even classes? e.g. for complex-valued signals?

Signals

- **Symmetry & Conjugate symmetry**
  - Are they closed under addition operation, scalar multiplications?
  - Examples?
  - **Direct-sum** of odd signals and even signals? Let's discuss very briefly.

Signals

- **Some useful signals**
  - exponential signals
  - sinusoidal signals
  - step signals
  - impulse signals
  - rectangle signals
  - ramp signals

Bounded Signals

- **Bounded signals** are of the form:
  - Continuous-time : $x(t)$, if there exists an M such that $|x(t)| \leq M$ for all t.
    - e.g. $x(t) = 3\sin 10t + 5\cos 6t$.
  - Discrete-time: $x[n]$, if an M such that $|x[n]| \leq M$ for all n.
    - e.g. $x[n] = \sin 10n - 5$.
  - How useful?

Periodic Signals

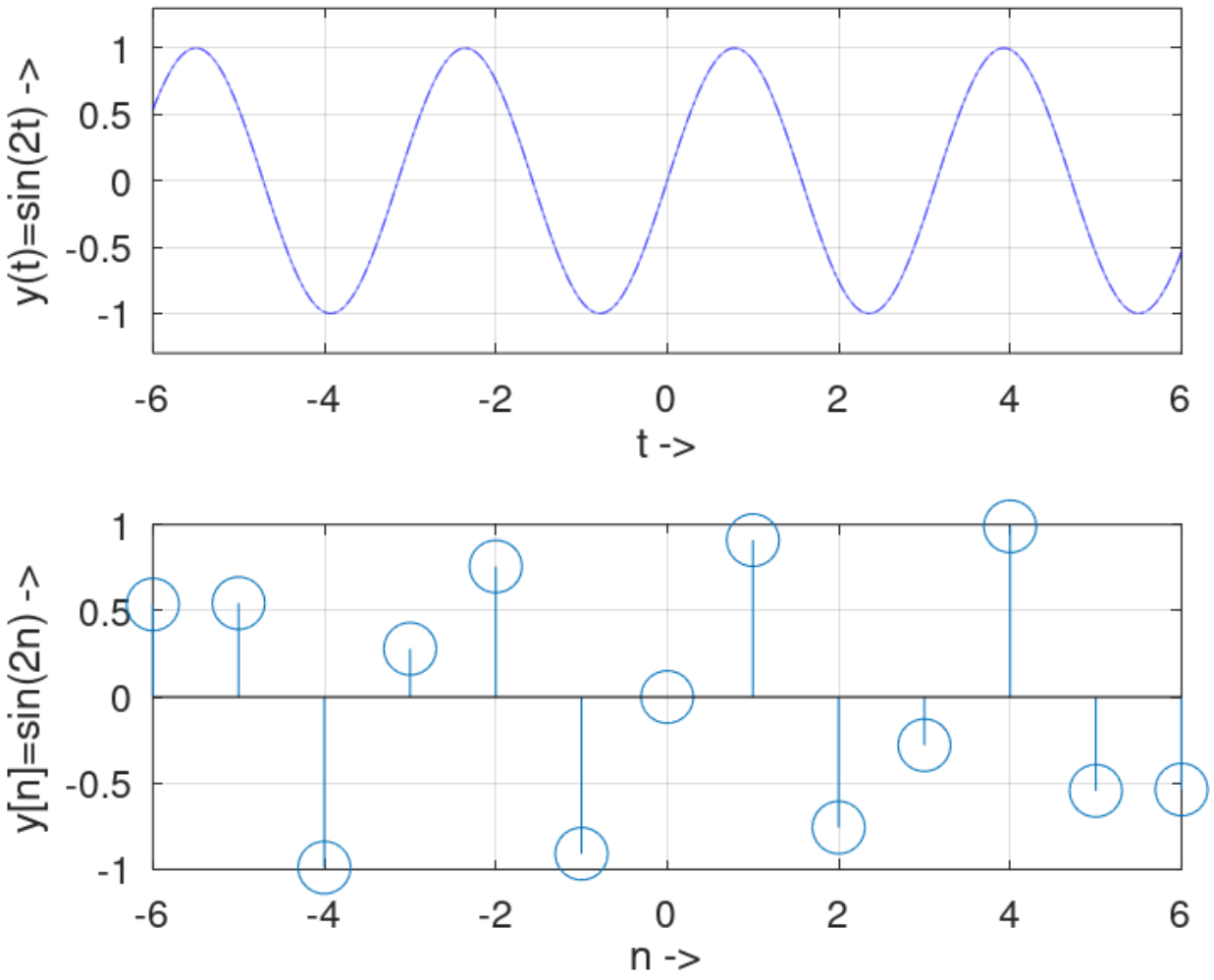
- **Periodic signals** are of the form:
  - Continuous-time: $x(t)$, if there exists a positive real T such that $x(t) = x(t+T)$.
    - $x(t)$ has a period T .
    - e.g. $x(t) = 3\sin 10t$.
  - Discrete-time: $x[n]$, if there exists a positive integer N such that $x[n] = x[n+N]$.
    - $x(t)$ has a period N .
    - e.g. $x[n] = n - 11 \cdot \text{floor}\left(\frac{n}{11}\right)$.
  - What about $x(t) = 3\sin 10t - 5\cos 7t$? What about $x[n] = 2\sin(10n+5)$?

Periodic Signals

- **Fundamental period** - Smallest period of a periodic signal.
- If the period of a periodic signal is T, then kT for all positive integer k is also a period.
- Signals that not periodic are called **aperiodic signals**.

Periodic Signals

- Are these periodic?



Exponential Signals

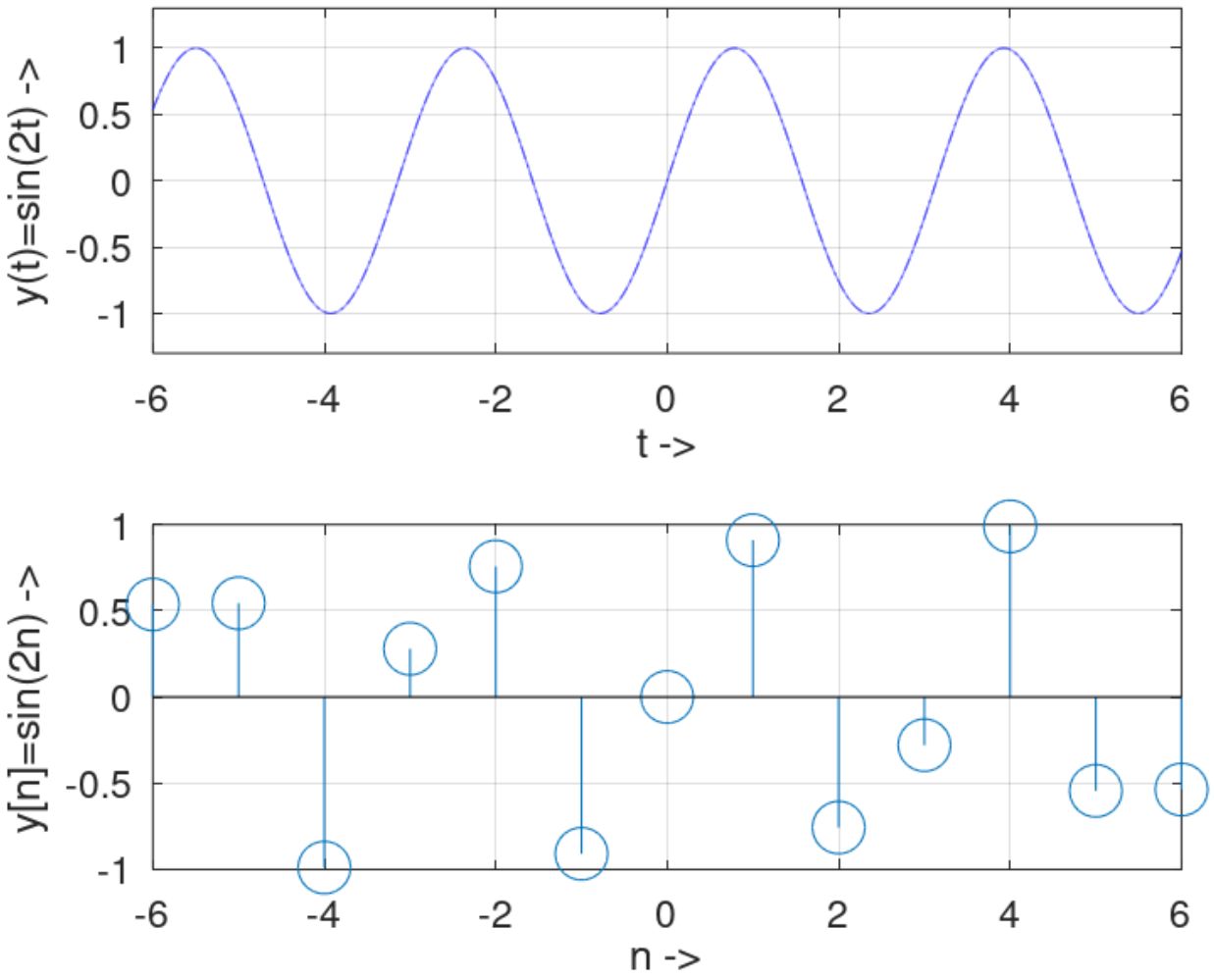
- Exponential signals are of the form:
  - Continuous-time : $x(t) = Ae^{bt} = Ac^t$ (very general)
    - If both A and b are real, then we have continuous-time **real exponential**.
  - Discrete-time: $x[n] = Ae^{bn} = Ac^n$ (very general)
    - If both A and b are real, then we have discrete-time **real exponential**.
  - How useful? Bounded? Periodic?

Sinusoidal Signals

- **Real sinusoidal signals** are of the form :
  - Continuous-time : $x(t) = A\cos(\omega t + \delta)$ and A, ω, δ are reals.
    - $x_1(t) = 3\sin(2t + 1)$.
  - Discrete-time: $x[n] = A\cos(\Omega n + \delta)$ and A, Ω, δ are reals.
    - $x_1[n] = 10\cos(5n + 4)$.
  - How useful? Bounded? Periodic?

Sinusoidal Signals

- **Real sinusoidal signals**:



Sinusoidal Signals

- **Complex sinusoidal signals** are of the form :
  - Continuous-time : $x(t) = Ae^{bt}$ and $b = j\omega$ is imaginary for some real ω.
    - $x(t) = Ae^{j\omega t} = \left(|A|\, e^{j\delta}\right) e^{j\omega t} = |A| \left(e^{j\delta} \cdot e^{j\omega t}\right) = |A| \left(\cos\left(\omega t + \delta\right) + j \sin\left(\omega t + \delta\right)\right)$.
  - Discrete-time: $x[n] = Ae^{bn} = Ac^n$ and $b = j\Omega$ is imaginary for some real Ω.
    - $x[n] = Ae^{j\Omega n} = \left(|A|\, e^{j\delta}\right) e^{j\Omega n} = |A| \left(e^{j\delta} \cdot e^{j\Omega n}\right) = |A| \left(\cos\left(\Omega n + \delta\right) + j \sin\left(\Omega n + \delta\right)\right)$.
  - How useful? Have we come across the above?
  - Bounded? Periodic?

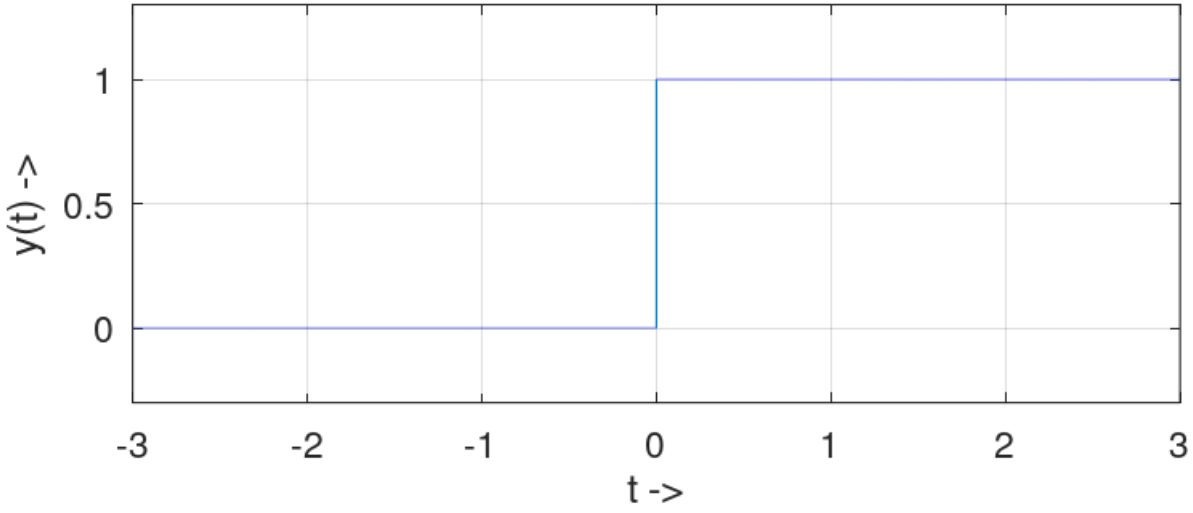
Exponential Signals

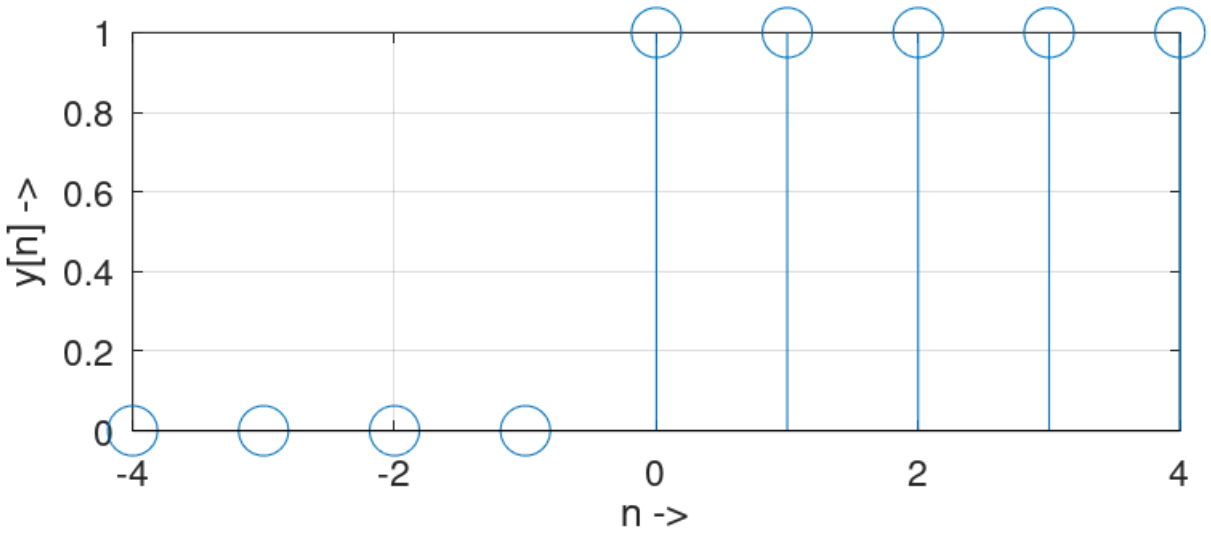
- **Continuous-time exponential signals**:
  - $x(t) = Ae^{bt}$ (very general)
  - Setting $A = |A|\, e^{j\delta}$ and $b = \sigma + j\omega$ $\quad (|A|, \delta, \sigma, \omega$ reals$)$
- $x(t) = |A|\, e^{(j\delta + \sigma t + j\omega t)} = \underbrace{|A|\, e^{\sigma t}}_{\text{real exponential}} \cdot \underbrace{e^{j(\omega t + \delta)}}_{\text{complex sinusoidal}}$.
- How useful? Bounded? Periodic?

Unit Step Signals

- **Continuous-time unit step signal** (Heaviside function):
  - $u(t) = \left\{ \begin{array}{ll} 1, & t \geq 0 \\ 0, & \text{otherwise} \end{array} \right\}$
- **Discrete-time unit step signal**:
  - $u[n] = \left\{ \begin{array}{ll} 1, & n \geq 0 \\ 0, & \text{otherwise} \end{array} \right\}$

Unit Step Signals





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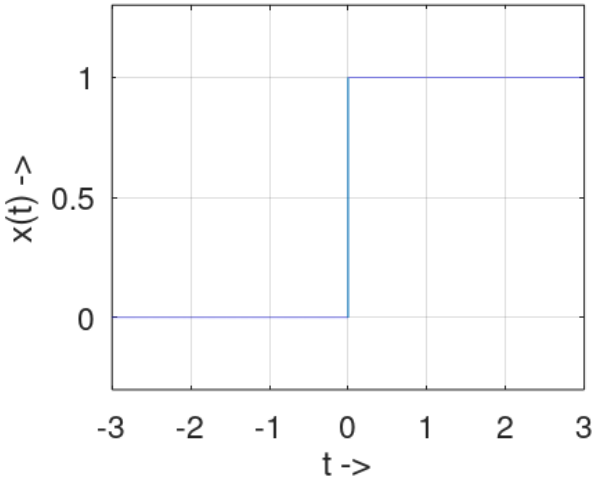
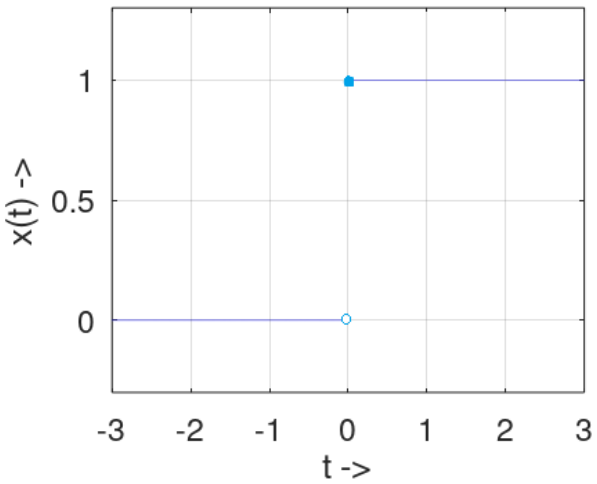
Signals

- Which one to consider?

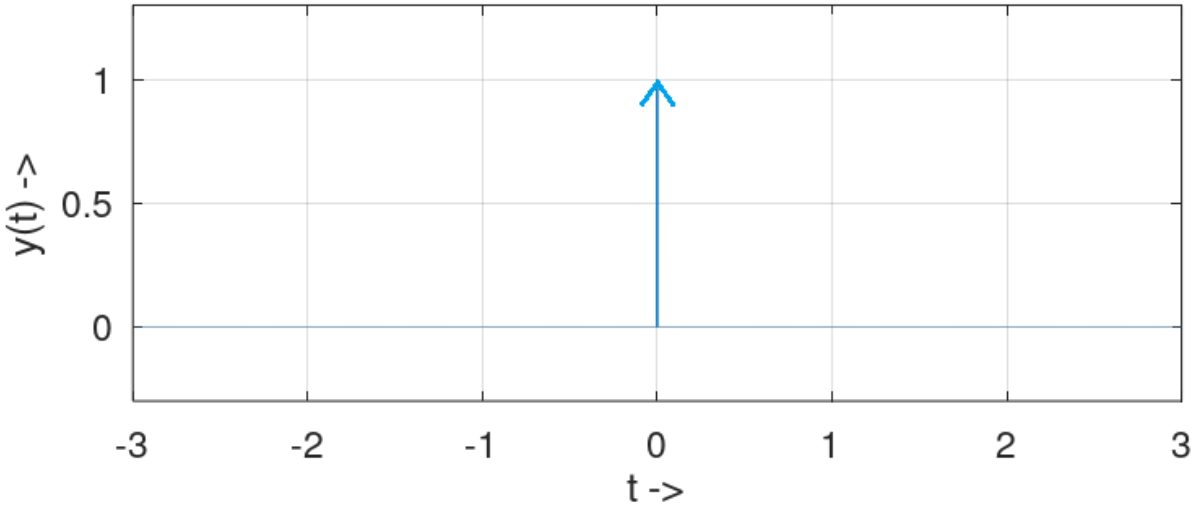


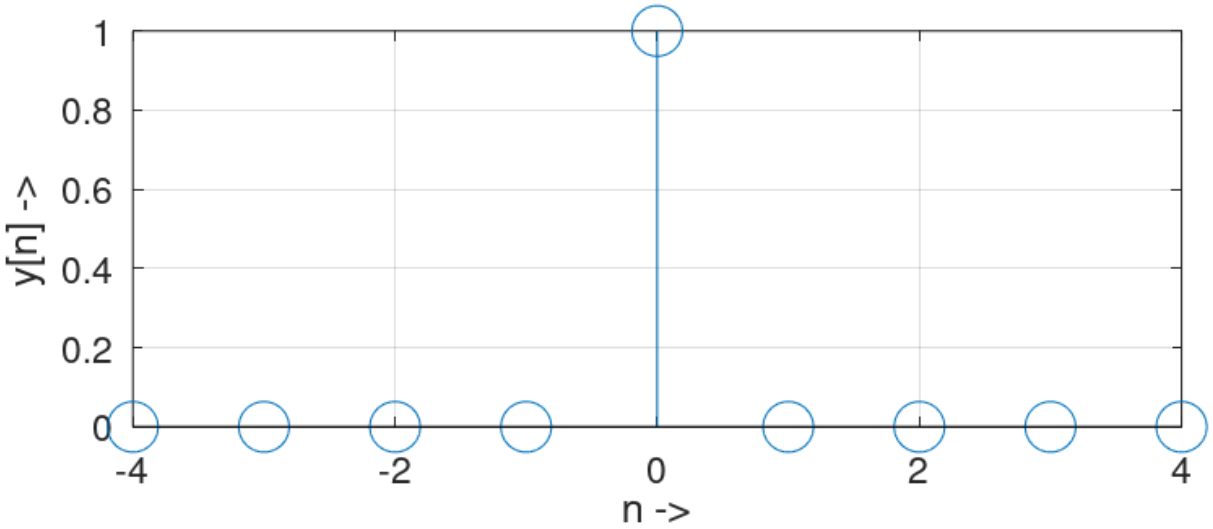


Unit Impulse Signals

- **Continuous-time unit impulse signal** (Dirac delta function):
  - $\delta(t)$ such that
    - $\delta(t) = 0$ for $t \neq 0$
    - $\int_{-\infty}^{\infty} \delta(t)dt = 1.$
  - not a usual function, but a generalised function (an arrow is used in the plot).
- **Discrete-time unit impulse signal**:
  - $\delta[n] = \left\{ \begin{array}{ll} 1, & n = 0 \\ 0, & \text{otherwise} \end{array} \right\}.$

Unit Impulse Signals





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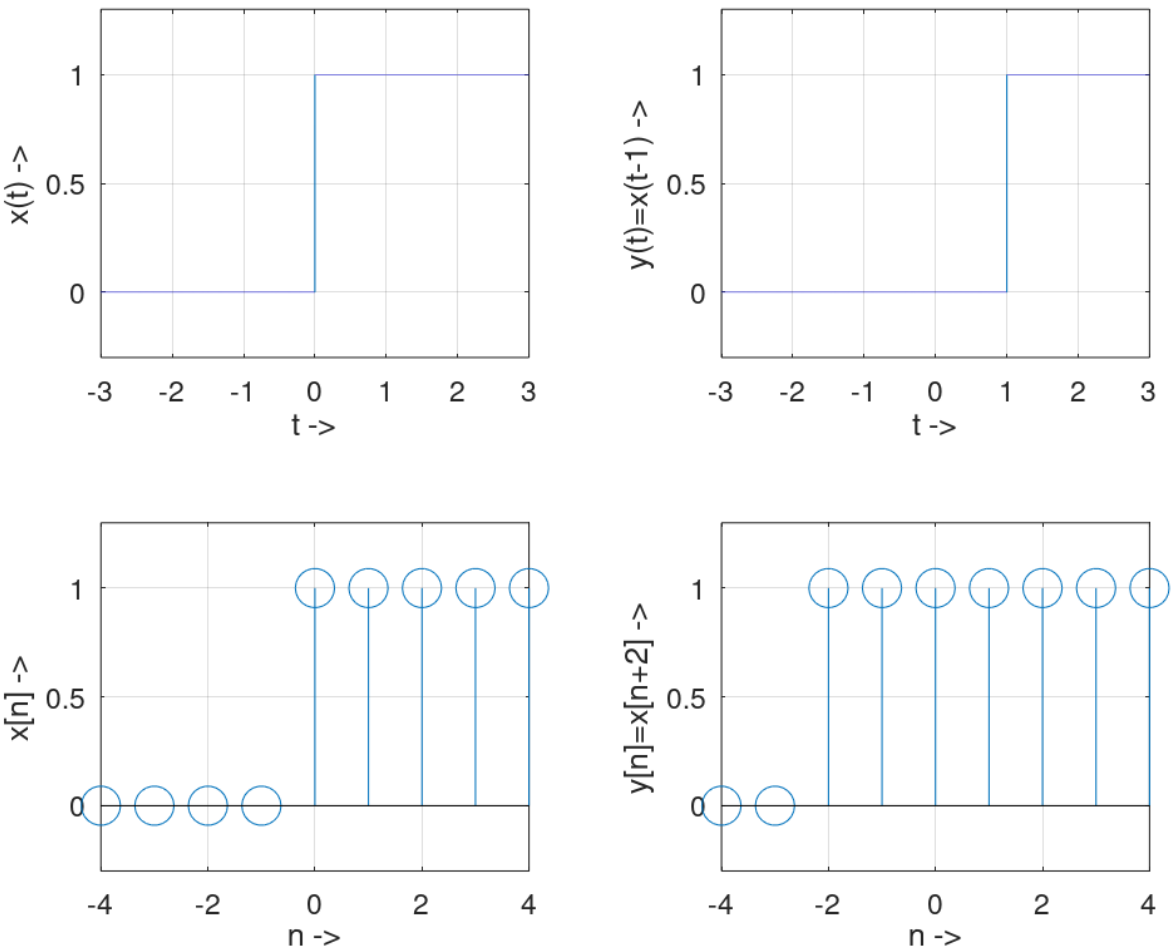
Transformations of Signals

- **Time-shifting**
  - Given a continuous-time signal $x(t)$, we have $y(t) = x(t - a)$ for a given real a.
  - Given a discrete-time signal $x[n]$, we have $y[n] = x[n - b]$ for a given integer b.

- When will the signals shift towards left? Or, towards right?
- Delayed or advanced?

Transformations of Signals

- **Time-shifting**

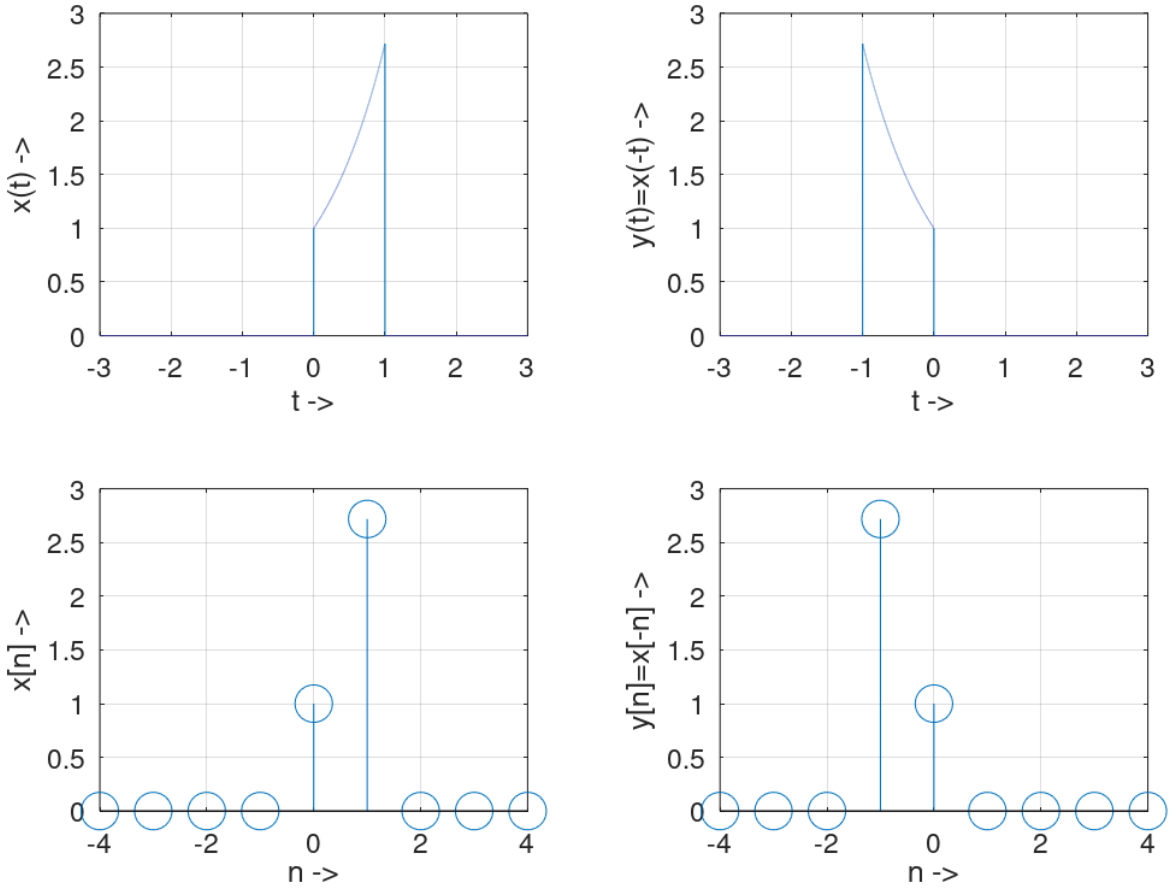


Transformations of Signals

- **Time-reversal**
  - Given a continuous-time signal $x(t)$, we have $y(t) = x(-t)$.
  - Given a discrete-time signal $x[n]$, we have $y[n] = x[-n]$.

Transformations of Signals

- **Time-reversal**



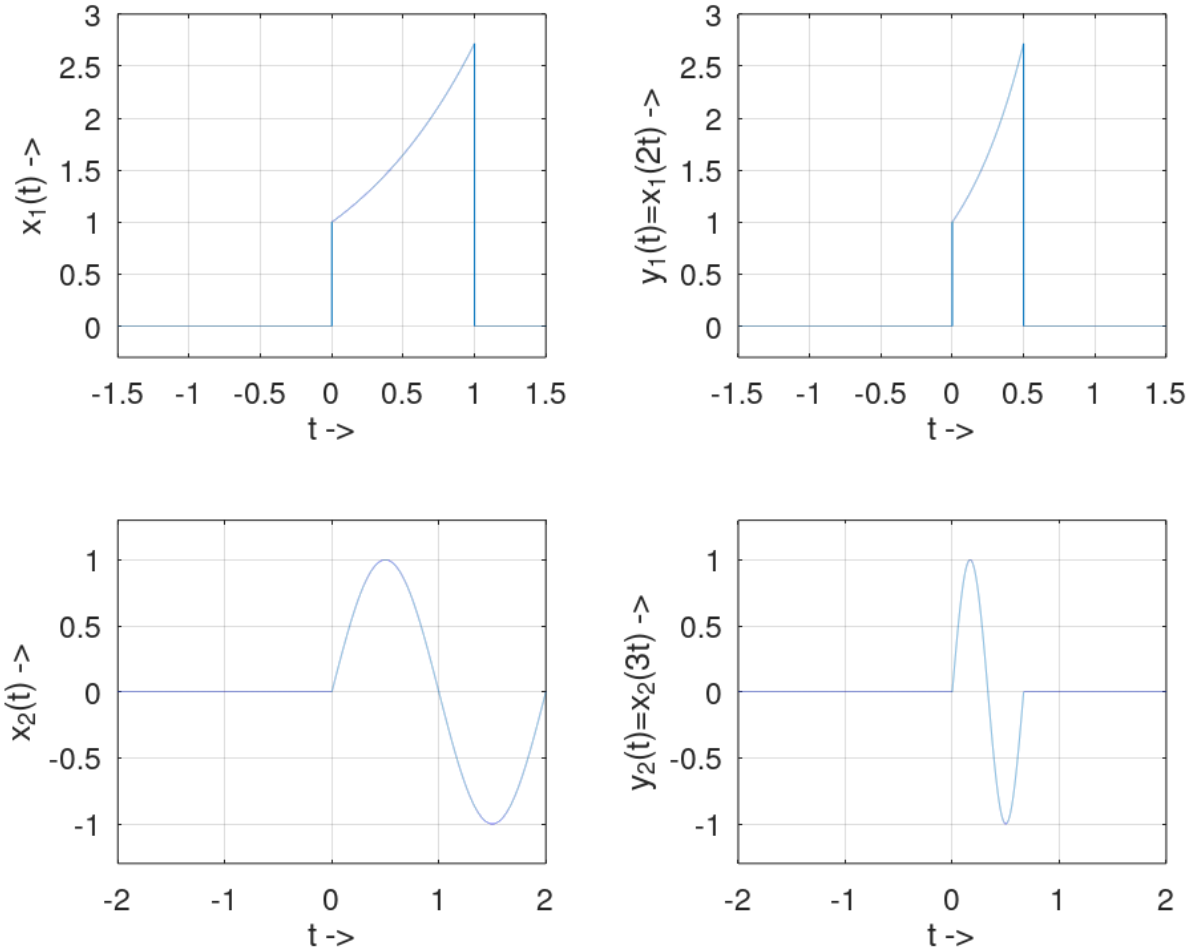
Transformations of Signals

- **Time-scaling**
  - Given a continuous-time signal $x(t)$, we have $y(t) = x(at)$ for a given non-zero real positive a.
  - Signals be expanded: when $a < 1$.
  - Compressed: when $a > 1$.

  - Given a discrete-time signal $x[n]$, does this make sense: $y[n] = x[bn]$ for a given real positive b (?).
  - We will see later downsampling and upsampling in discrete-time domain.

Transformations of Signals

- **Time-scaling**



Unit Impulse Signals - Properties

- **Continuous-time unit impulse signal** (Dirac delta function):
  - $\delta(t)$ such that
    - $\delta(t) = 0$ for $t \neq 0$, $\displaystyle\int_{-\infty}^{\infty} \delta(t)dt = 1$.
  - Consider a sequence of **rectangular pulse** (at time $t = 0$) functions
    - $q_n(t) := \begin{cases} 0 & t \leq -\dfrac{1}{2n} \\ n & -\dfrac{1}{2n} < t < -\dfrac{1}{2n} \\ 0 & t \geq \dfrac{1}{2n} \end{cases}$.

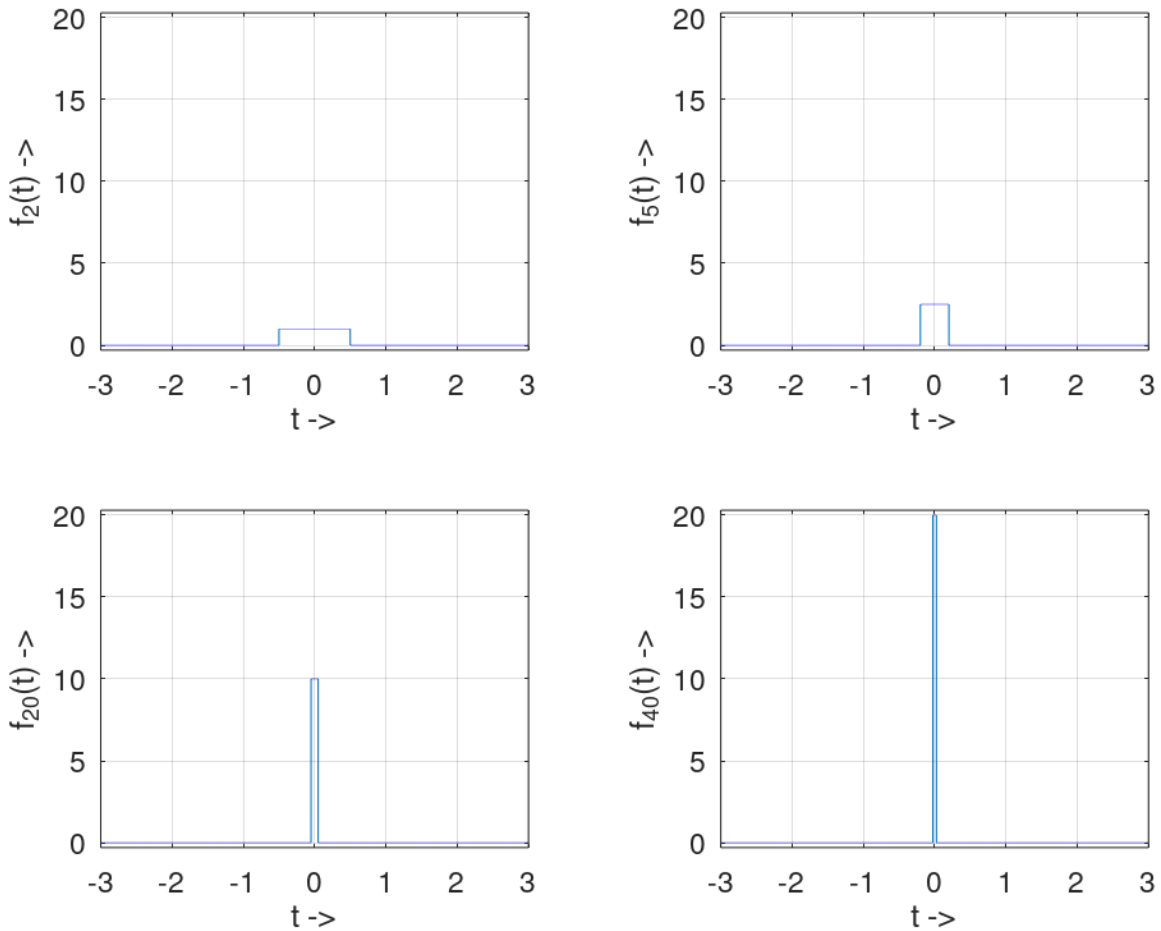
Unit Impulse Signals - Properties

- For all finite n, we have

$$\begin{aligned}
\int_{-\infty}^{\infty} \boldsymbol{q_n(t)} dt &= \int_{-\infty}^{-\frac{1}{2n}} q_n(t)dt + \int_{-\frac{1}{2n}}^{+\frac{1}{2n}} q_n(t)dt + \int_{+\frac{1}{2n}}^{\infty} q_n(t)dt \\
&= \int_{-\infty}^{-\frac{1}{2n}} \mathbf{0} \cdot dt + \int_{-\frac{1}{2n}}^{+\frac{1}{2n}} \boldsymbol{n} \cdot dt + \int_{+\frac{1}{2n}}^{\infty} \mathbf{0} \cdot dt \\
&= 0 + \boldsymbol{n} \int_{-\frac{1}{2n}}^{+\frac{1}{2n}} \boldsymbol{dt} + 0 \\
&= n \cdot \frac{1}{n} = 1.
\end{aligned}$$

  - If $n \to \infty$, what happens? What if functions other than **rectangular pulse**?.

Unit Impulse Signals - Properties



- If $n \to \infty$, what happens? What if functions other than **rectangular pulse**?

Unit Impulse Signals - Properties

- **Continuous-time unit impulse signal** (Dirac delta function):
  - $\delta(t)$ such that
    - $\delta(t) = 0$ for $t \neq 0$, $\displaystyle\int_{-\infty}^{\infty} \delta(t)dt = 1$.
  - What about a given function f continuous at $t = 0$?

We have

$$\begin{aligned}
\int_{-\infty}^{\infty} f(t)\delta(t)dt &= \int_{-\infty}^{0^-} f(t)\delta(t)dt + \int_{0^-}^{0^+} f(t)\delta(t)dt + \int_{0^+}^{\infty} f(t)\delta(t)dt \\
&= \int_{-\infty}^{0^-} f(t) \cdot \mathbf{0} \cdot dt + \int_{0^-}^{0^+} f(t)\delta(t)dt + \int_{0^+}^{\infty} f(t) \cdot \mathbf{0} \cdot dt \\
&= 0 + \boldsymbol{f(0)} \int_{\mathbf{0}^-}^{\mathbf{0}^+} \boldsymbol{\delta(t)dt} + 0 \\
&= f(0).
\end{aligned}$$

Unit Impulse Signals - Properties

- What about a given function f continuous at $t = t_0$?

$$
\begin{aligned}
& \int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt \\
= & \int_{-\infty}^{t_0^-} f(t)\delta(t-t_0)dt + \int_{t_0^-}^{t_0^+} f(t)\delta(t-t_0)dt + \int_{t_0^+}^{\infty} f(t)\delta(t-t_0)dt \\
= & \int_{-\infty}^{t_0^-} f(t) \cdot \mathbf{0} \cdot dt + \int_{t_0^-}^{t_0^+} f(t)\delta(t-t_0)dt + \int_{t_0^+}^{\infty} f(t) \cdot \mathbf{0} \cdot dt \\
= & \; 0 + \boldsymbol{f(t_0)} \int_{t_0^-}^{t_0^+} \boldsymbol{\delta(t-t_0)dt} + 0 \\
= & \; f(t_0).
\end{aligned}
$$

- This is called **sifting/sampling property** of $\delta(t)$.

Unit Impulse Signals - Properties

- Also, for any function continuous at $t = t_0$, we have

$$f(t)\delta(t - t_0) \quad = \quad f(t_0)\delta(t - t_0)$$

- This is called the **equivalence property** of $\delta(t)$.
- What about this?
  - Define $g(s) := \int_{-\infty}^{\infty} f(s)\delta(t - s)dt$.

Unit Impulse Signals - Properties

- **Discrete-time unit impulse signal**:
  - $\delta[n] = \left\{ \begin{array}{ll} 1, & n = 0 \\ 0, & \text{otherwise} \end{array} \right\}$.
- Similar to continuous-time domain, we have
  - $\sum_{n=-\infty}^{\infty} f[n]\delta[n] = f[0]$
  - $\sum_{n=-\infty}^{\infty} f[n]\delta[n-n_0] = f[n_0]$ for all integer n_0(**sifting/sampling property**)
  - $f[n]\delta[n-n_0] = f[n_0]\delta[n-n_0]$ (**equivalence property**).

Unit Impulse Signals - Properties

- **Continuous-time unit impulse signal** (Dirac delta function):
  - $\delta(t)$ such that $\delta(t) = 0$ for $t \neq 0$, $\int_{-\infty}^{\infty} \delta(t)dt = 1$.
- Reconsider a sequence of **rectangular pulse** (at time $t = 0$) functions
  - $q_n(t) := \begin{cases} 0, & t \leq -\frac{1}{2n} \\ n, & -\frac{1}{2n} < t < -\frac{1}{2n} \\ 0, & t \geq \frac{1}{2n} \end{cases}$.

Unit Impulse Signals - Properties

- Reconsider the integration of $q_n(t)$:

$$\int_{-\infty}^{t} q_n(t)dt = \begin{cases} \int_{-\infty}^{t} \mathbf{0}dt, & t \leq -\frac{1}{2n} \\ \int_{-\infty}^{-\frac{1}{2n}} \mathbf{0} \cdot dt + \int_{-\frac{1}{2n}}^{t} \boldsymbol{n}dt, & -\frac{1}{2n} < t < -\frac{1}{2n} \\ \int_{-\infty}^{-\frac{1}{2n}} \mathbf{0} \cdot dt + + \int_{-\frac{1}{2n}}^{+\frac{1}{2n}} \boldsymbol{n}dt \int_{+\frac{1}{2n}}^{t} \mathbf{0} \cdot dt, & t \geq \frac{1}{2n}. \end{cases}$$

$$= \begin{cases} \mathbf{0}, & t \leq -\frac{1}{2n} \\ \boldsymbol{n}(t + \frac{1}{2n}), & -\frac{1}{2n} < t < -\frac{1}{2n} \\ \mathbf{1}, & t \geq \frac{1}{2n}. \end{cases}.$$

Unit Impulse Signals - Properties

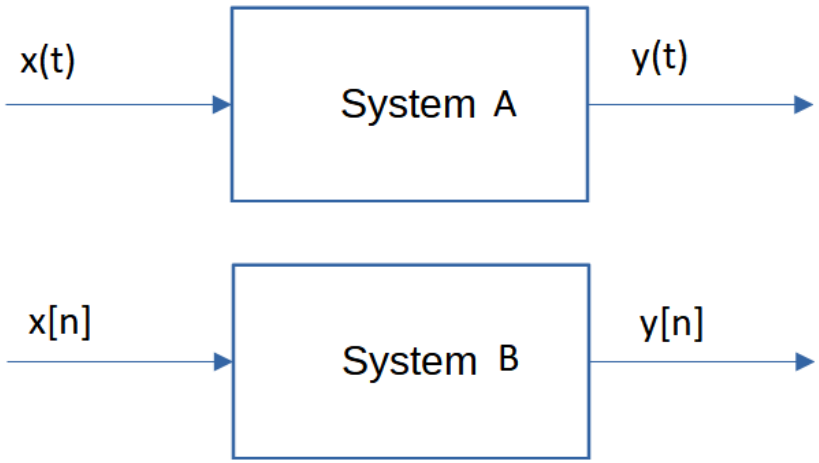
$$\int_{-\infty}^{t} q_n(t)dt = \begin{cases} \mathbf{0}, & t \leq -\frac{1}{2n} \\ \boldsymbol{n}(t + \frac{1}{2n}), & -\frac{1}{2n} < t < -\frac{1}{2n} \\ \mathbf{1}, & t \geq \frac{1}{2n}. \end{cases}.$$

- What as $n \to \infty$?
- How about the step function $u(t)$ with $u(0) = \frac{1}{2}$?

Systems

- **Systems**
  - are mathematical entities (**functions/process**) of a number of **input signals** and a number of **output signals**. We write, in general, $y(t) = \mathcal{H}_1(x(t))$, $y[n] = \mathcal{H}_2(x[n])$.
  - behave as what are called **'functionals'** i.e. map **functions** (signals) to **functions** (signals).
  - Simple examples: $y(t) = 10x(t)$, $y[n] = 10x[n]$

Here $x(t)$ and $x[n]$ are the input signals and $y(t) := \mathcal{A}(x(t))$ and $y[n] := \mathcal{B}(x[n])$ are the output signals.



Systems

- **Classification of Systems**
  - **Continuous-time** systems: Where both input and output signals are continuous-time signals.
  - **Discrete-time** systems: Where both input and output signals are discrete-time signals.
  - **Mixed** discrete-continuous systems.
  - Other systems with non-time variable signals are generalised versions.

Systems

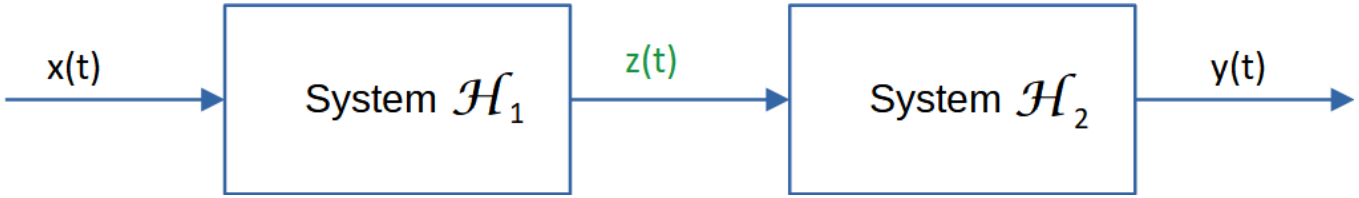
- **Classification of Systems**
  - Input signals
    - System with one input signal - **Single Input (SI)** system
    - System with multiple input signals - **Multiple Input (MI)** system
  - Output signals
    - System with one output signal - **Single Output (SO)** system
    - System with multiple output signals - **Multiple Output (MO)** system
  - e.g. SISO system, MISO system, MIMO system, etc.

Systems

- **Systems - examples**
  - Communication systems
    - signals - message
    - systems - transmitter, channel, receiver
  - Electrical systems
    - signals - voltages, currents,
    - systems - portions of circuits with input sources and output voltages/currents
  - Control systems
    - signals - reference inputs, sensor outputs, position, velocity, temperature
    - systems - controller, plant
  - Signal conditioning systems
    - signals - audio, video
    - systems - filters
  - Mechanical systems, Acoustic systems, etc.

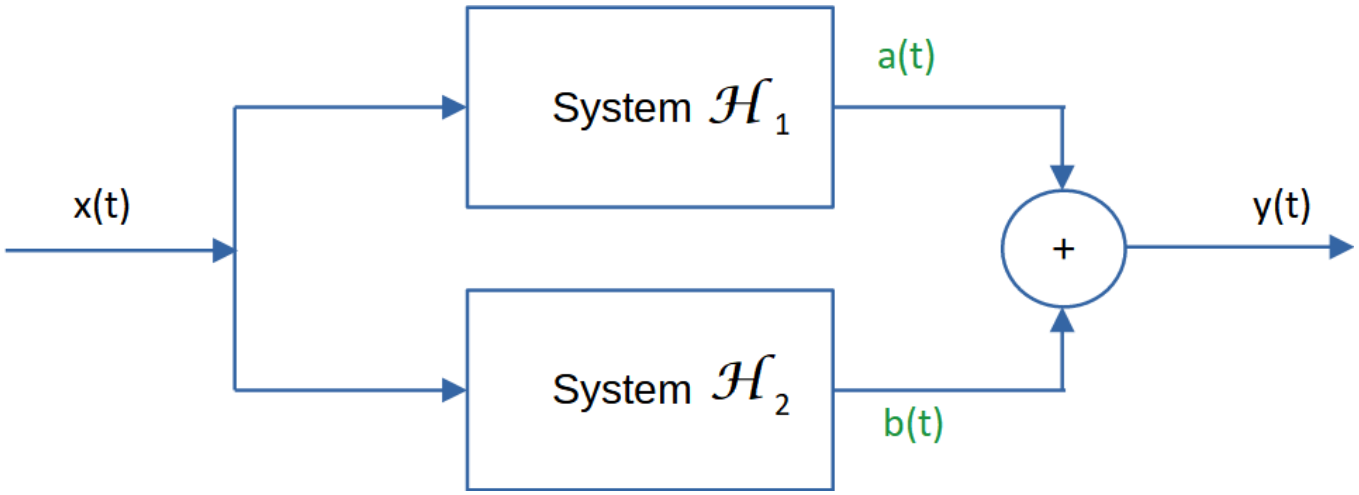
Interconnection of Systems

Series/Cascaded Connection:



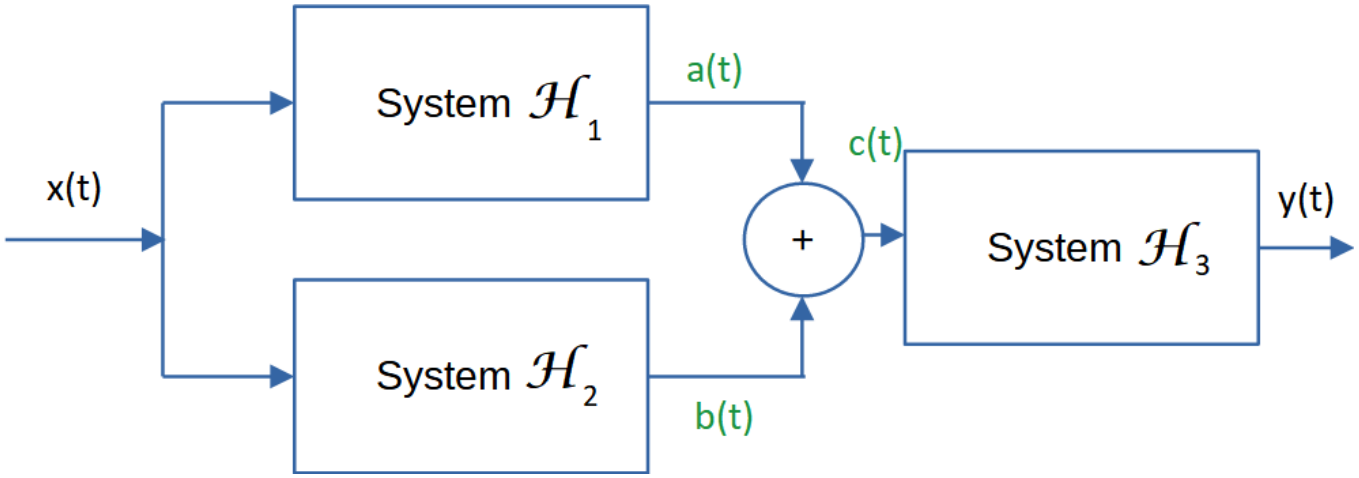
- We have
  - $z(t) = \mathcal{H}_1(x(t))$ and $y(t) = \mathcal{H}_2(z(t))$,
  - $\boldsymbol{y(t) = \mathcal{H}_2(\mathcal{H}_1(x(t)))}$.

Parallel Connection:



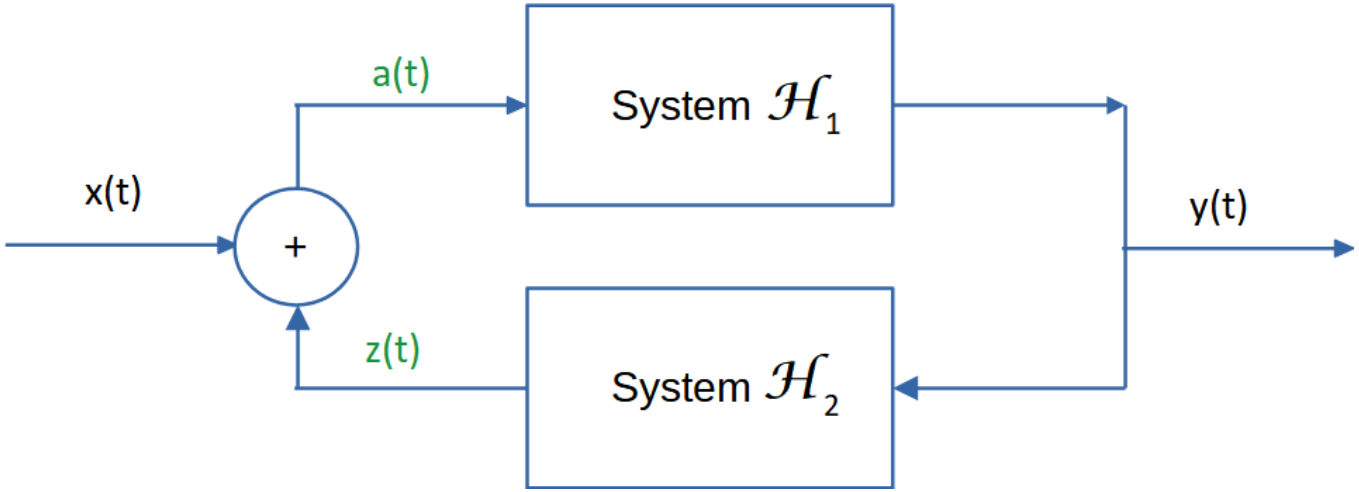
- We have
  - $a(t) = \mathcal{H}_1(x(t))$ and $b(t) = \mathcal{H}_2(x(t))$.
  - $\boldsymbol{y(t)} = a(t) + b(t) = \boldsymbol{\mathcal{H}_1(x(t)) + \mathcal{H}_2(x(t))}$.

Combined Parallel-Series Connection:



- We have
  - $a(t) = \mathcal{H}_1(x(t))$ and $b(t) = \mathcal{H}_2(x(t))$.
  - $c(t) = a(t) + b(t) = \mathcal{H}_1(x(t)) + \mathcal{H}_2(x(t))$
  - $\boldsymbol{y(t)} = \mathcal{H}_3(c(t)) = \boldsymbol{\mathcal{H}_3(\mathcal{H}_1(x(t)) + \mathcal{H}_2(x(t)))}$.

Feedback Connection:



- We have
  - $a(t) = x(t) + z(t)$, $y(t) = \mathcal{H}_1(a(t))$ and $z(t) = \mathcal{H}_2(y(t))$.
  - $\boldsymbol{y(t)} = \mathcal{H}_1(a(t)) = \mathcal{H}_1(x(t) + z(t)) = \boldsymbol{\mathcal{H}_1(x(t) + \mathcal{H}_2(y(t)))}$.
  - And, then?

Systems - Memory

- **Memoryless Systems:**
  - If output at each independent domain value is **dependent only** on the input at the same domain value.
    - e.g. Amplification: $y(t) = 10x(t)$
    - How about $y(t) = 2x(t) - sin^2(t+1)$?
- **Systems with Memory**:
  - If output at each independent domain value is dependent on the input at any other domain values.
    - e.g. Accumulation: $y(t) = \int_{-\infty}^{t} x(t)dt$
    - How about $y(t) = 2x(t-1) - sin^2(t)$?

Systems - Causality

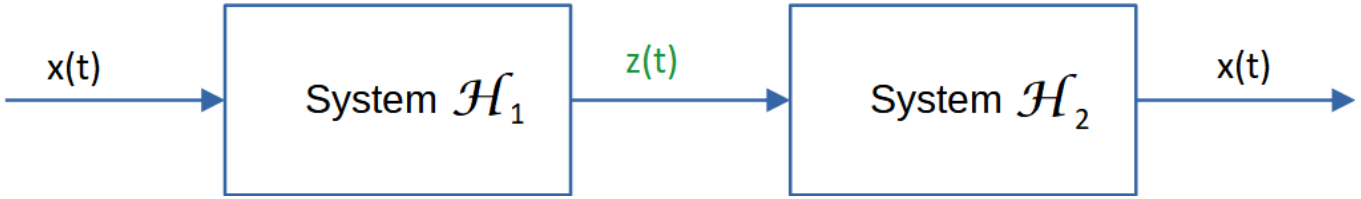
- **Causal/Non-anticipatory Systems:**
  - If output at each independent domain value is **dependent only** on the input at the same domain value and previous domain values.
    - e.g. $y(t) = 10x(t)$, $y(t) = 2x(t) - x^2(t-2)$
    - How about $y(t) = y(t+1) + x(t)$?
- **Non-Causal/Anticipatory Systems:**:
  - If output at any independent domain value is dependent on the input at any future domain value.
    - e.g. $y(t) = \int_{-\infty}^{t} x(-t)dt$
    - How about $y(t) = 2x(t+1) - x^2(t-2)$?
    - Realisable?
- **[BTW]** A **signal** $x(t)$ is said to be **causal** if it has zero for all $t < 0$. (Similar for discrete-time $x[n]$.)

Systems - Stability

- **Bounded-Input Bounded-Output (BIBO) Stability:**
  - BIBO stable, if every **bounded input** produces a **bounded output**.
  - Continuous-time systems:
    - i.e. for any input $x(t)$ with $|x(t)| \leq B_x$ finite, output $y(t)$ has $|y(t)| \leq B_y$.
  - In other words, roughly, small inputs do not produce arbitrarily large outputs.
- **Uniformly-BIBO Stability:**
  - Bound B_y is independent of $x(t)$.
- We will look in details later.

Systems - Invertibility

- **Invertible Systems:**
  - invertible, if there exists another system such that the cascased combination leads to the identity system.
  - Continuous-time systems:



  - We have
    - $x(t) = \mathcal{H}_1(x(t))$ and $y(t) = \mathcal{H}_2(x(t))$,
    - $\boldsymbol{x(t) = \mathcal{H}_2(\mathcal{H}_1(x(t)))}$.
    - We denote $\mathcal{H}_2 = \mathcal{H}_1^{-1}$.